Scaling Our Database Layer

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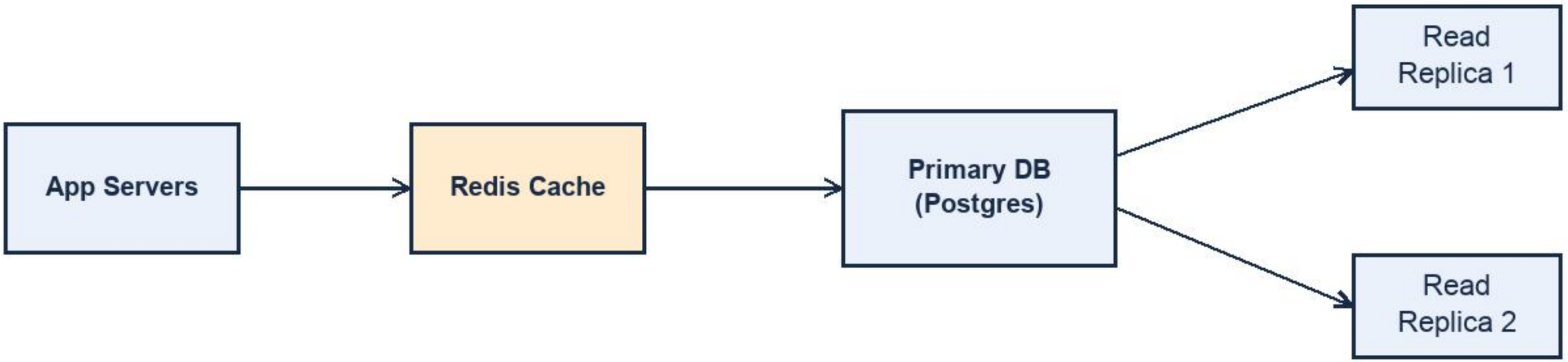
The Problem: Database Overload

- Checkout API p99 latency up 4x during peak
- Primary Postgres DB at 95% CPU at peak hours
- Every request hits the database directly (no cache)



Proposed Architecture: Cache + Read Replicas

- Redis cache in front of DB absorbs hot reads
- Two read replicas handle reporting queries
- Writes still go to primary; async replication



Impact

- Primary DB load down 70%
- p99 latency back to baseline
- Read replicas absorb nearly all reporting traffic

Next Steps

- Finalize cache invalidation strategy
- Add cache hit-rate monitoring dashboards
- Document on-call runbook for new architecture